

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

Network: Stellar testnet (mainnet is a later phase).

Scope: The sealed-auction integration kit only (public receipt + template + 3 live rounds).

Regulation Posture: Non-custodial escrow, no yield.

Drafting Method: Drafted with an adversarial AI persona review method (not a team).

Field	Details
Project Name	Sub Rosa
Builder / Team Name	Emin Karagoz
Primary Contact (Name + Email)	Emin Karagoz · eminkaragoz07@gmail.com
Ambassador Chapter	Stellar Türkiye
Ambassador Chapter Lead	İrem Koçi (Stellar Türkiye)
Date Submitted	2026-07-12
Suggested Sprint Start Date	2026-07-15

Supporting Links (Existing Work)

- GitHub repo: <https://github.com/karagozemin/Sub-Rosa>
 - Live demo: <https://sub-rosa-web.vercel.app>
 - Demo video: <https://youtu.be/NDuR5B2ztQo>
 - Architecture: <https://github.com/karagozemin/Sub-Rosa/blob/main/ARCHITECTURE.md>
 - Pitch deck: <https://gamma.app/docs/SUB-ROSA-g8o9ulvc7nezz4j?mode=doc>
-

2. Instawards Overview & Intent

2.1 Instawards Purpose (for Builder Context)

Instawards support short, clearly scoped, execution-focused work that helps a project make tangible progress toward building on Stellar — specific, achievable outcomes that can be completed and demonstrated within 30 days or less. This SOW is a shared commitment

between the Builder and the Ambassador Chapter Lead on what will be delivered, why it matters, and how success will be verified.

Section	Description
Problem Being Addressed	Sealed-bid auctions need bids to stay secret until a fair reveal, and they need every bidder to trust that losers get refunded and the winner actually pays. Doing this on-chain normally means writing timelock cryptography, an escrow contract, and a keeper that reveals and settles on time — most builders never get past that wall. Sub Rosa already solved it: escrow-backed sealed-auction infrastructure on Stellar that won 1st Place in the Hack Privacy Track at Build On Stellar (IBW 2026). But the code being proven is not the same as it being usable. Today an outside builder still cannot verify an auction result from a single public link, and cannot run their own round from the docs. That gap is exactly what blocks real adoption.
Objective of This Instaward	After 30 days, Sub Rosa will have a live, end-to-end sealed-auction integration kit on Stellar testnet: any observer can verify a settled auction from one public receipt link, and any builder can run their own round from a copy-paste template. Completion is provable by publicly verifiable transaction hashes on a testnet explorer. Success metric (binary): ≥ 3 verifiable, end-to-end testnet rounds — each a full commit $\rightarrow$ reveal $\rightarrow$ settle $\rightarrow$ refund cycle — with a public receipt link and on-chain settlement + refund transaction hashes anyone can look up on a testnet explorer. Pass = 3 receipts with matching tx hashes; there is no partial credit. Why this is achievable in 30 days: This plan does not start from zero. The hard technical risks are already retired by Sub Rosa's hackathon-winning build — a working Soroban round contract, Drand-based timelock encryption (tlock), an escrow + refund flow, and an automated keeper that reveals and settles — with a passing test suite. The 30 days are about packaging proven parts into one public, click-to-verify kit, not discovering whether it can be done.

4. Scope of Work (30-Day Deliverables)

4.1 In-Scope Deliverables

Important guidance: This scope must be achievable within **30 calendar days**. If the work feels larger, it should be reduced or split into more achievable phases.

Deliverable	Description (what will be built or produced)	Why this matters
D1: Public auction receipt (page + SDK module)	An SDK module and public receipt page that turn any settled round into a single verifiable record: round ID, contract ID, Drand round, bidder count, escrow amount, winning bid, settlement + refund transactions, and Stellar Expert links. Covered by tests proving conservation — refunds plus operator payment always equal the escrow — so a reader can trust the numbers without reading the contract.	Anyone can open one link and confirm the auction settled fairly, with no code reading needed. This is the trust surface the whole kit rests on.
D2: Integration template + quickstart	A copy-paste template and step-by-step quickstart (install → create a round → commit a sealed bid → reveal with Drand → settle and refund) so an outside builder can run their own sealed auction on testnet without writing the timelock crypto, keeper, or escrow logic themselves. Driven end to end to create one real testnet round through the template.	Proves Sub Rosa is reusable infrastructure, not a one-off demo, and makes it trivial for a pilot partner to try.
D3: Live testnet demo + 3 published receipts	Deploy the demo publicly and run 3 sealed-auction rounds (3–10 bidders each) using XLM or USDC escrow, each a full commit → reveal → settle → refund cycle. Publish 3 full public receipts with settlement and refund transaction links, a public repo, and a short evidence pack (screenshots or a clip + a plain-language write-up of the lifecycle and known limits).	Creates real Stellar testnet activity and gives repeatable, independently verifiable proof of the full auction lifecycle — tx hashes, not just a screen recording.

Out of Scope (Explicitly Not Included)

- **Mainnet / real value** — testnet only; no real-value rounds this sprint.

- **Third-party security audit** — no external audit of the contracts or keeper.
- **Confirmed named pilot partner** — partner outreach is best-effort, not a required deliverable.
- **Production keeper infrastructure** — the keeper runs demo-grade; no HA deployment, alerting, or key-management/HSM setup.
- **New cryptographic primitives** — uses the existing Drand tlock and escrow design; no new crypto is designed or changed.
- **Governance / fee economics** — no operator-fee market, dispute resolution, or on-chain governance.
- **Production persistence & scale** — no production DB, multi-tenant operator dashboard, or high-throughput/anti-sybil hardening beyond demo needs.
- **Wallet / account UX beyond the demo** — no custom wallet, recovery, or onboarding flows outside the receipt and template pages.

4.2 Deliverable-Aligned Budget Request

Requested Budget Amount	Rationale for Budget Request
\$5,000 USDC	This Instaward funds execution, not exploration. The hard technical risks — sealed bidding, Drand timelock reveal, escrow, refunds, and an automated keeper — are already retired by Sub Rosa's hackathon-winning testnet build, so this 30-day scope is about packaging those proven building blocks into one public, verifiable integration kit, not researching whether it can be done. A 30-day solo builder stipend covers focused engineering across three shippable deliverables (public receipt page + SDK module, integration template with quickstart, and a live testnet demo), testnet operations (keeper runtime and transaction costs across 3 rounds), and production of the verifiable evidence pack. Tied directly to the three deliverables, each of which is publicly verifiable on completion. The amount is deliberately scoped to a single integration kit on testnet, with mainnet reserved for a later phase.

5. 30-Day Execution Plan & Timeline

5.1 Weekly Breakdown

Week	Planned Work	Expected Output
Week 1: Receipt module + conservation tests (de-risk the trust surface first)	Finish the SDK receipt module that reads a settled round on-chain and assembles the full record (round ID, contract ID, Drand round, bidder count, escrow, winning bid, settlement + refund txs, Stellar Expert links). Add the conservation tests (refunds + operator payment = escrow). Build the public receipt page rendering a real settled round.	Receipt module merged with passing conservation tests. Public receipt page showing one real settled testnet round with clickable, verifiable tx hashes. The trust surface is proven by end of Week 1.
Week 2: Integration template + first round through it	Build the copy-paste template and quickstart docs (install → create round → commit → reveal → settle/refund). Drive one real testnet round end to end through the template exactly as an outside builder would, using only the documented steps.	Public template and quickstart published. One testnet round created entirely through the template, with a receipt link proving it. D2 happy path works from docs alone.
Week 3: Live demo deployed + 3 rounds run	Deploy the demo to a public testnet URL. Run 3 sealed-auction rounds (3–10 bidders each) using XLM or USDC escrow, each a full commit → reveal → settle → refund cycle. Harden error handling and keeper config for repeatable	Live demo URL reachable. 3 completed rounds with on-chain settlement and refunds. Public repo published. D3 substantially complete.

	runs. Publish/clean the public repo.	
Week 4: Publish receipts + evidence pack	Publish the 3 full public receipts, gather screenshots or a short clip, write the plain-language lifecycle + limits write-up, tidy the docs, and produce the wrap-up report.	3 public receipt links + explorer tx hashes (settlement + refund per round). Evidence pack and short completion report published in the public repo. D1 + D2 + D3 closed; SOW deliverables shippable.

6. Evidence of Completion (Required)

6.1 Planned Evidence to Be Submitted

Important guidance: Evidence should be clear, verifiable, and easy to review by the Ambassador Chapter Lead **with minimal technical expertise**.

Deliverable	Evidence Type	Description
D1: Public auction receipt	Live link + public repo + test output	Click the receipt link to open a public page showing a settled round's status, escrow, winning bid, and Stellar Expert links, all loading from on-chain data. Open the linked repo to see the receipt module and run the passing conservation tests yourself.
D2: Integration template + quickstart	Public template (repo) + testnet tx (link)	Open the template and quickstart in the public repo, then click the transaction link to confirm that one testnet round was created through it on a public testnet explorer.
D3: Live demo + 3 receipts	Live demo URL + 3 receipt links (tx hashes) + demo video	Open the live demo URL, then click each of the 3 public receipt links to confirm settlement and refund transactions on the public testnet explorer, and watch the short demo video (youtu.be/NDuR5B2ztQo) showing the full commit → reveal → settle → refund lifecycle.

6.2 Evidence Verification Checklist (For Ambassador Use)

For each deliverable, the Ambassador Chapter Lead will assess whether evidence is present and sufficient.

Deliverable	Evidence Present	Evidence Partial	Evidence Missing	Comments
D1: Public auction receipt	☐	☐	☐	
D2: Integration template + quickstart	☐	☐	☐	
D3: Live demo + 3 receipts	☐	☐	☐	

Completion Addendum

Sub Rosa sealed-auction integration kit

PASS - SOW COMPLETED

The binary success metric is met: three independently verifiable end-to-end testnet Auction rounds, each with a public receipt and on-chain settlement/refund evidence.

PROJECT
Sub Rosa

BUILDER
Emin Karagoz

NETWORK
Stellar testnet

PUBLIC APP
sub-rosa.online

CORE V2 CONTRACT
CCOVGOQQZ...C5OV

PAYMENT / ESCROW
Native testnet XLM

Completion Checklist

Deliverable	Completed output	Evidence	Result
D1 Public auction receipt	SDK evidence module and public receipt page. Each page shows round ID, contract, Drand round, bidder count, escrow, winning bid, lifecycle hashes, and live-ledger reconciliation.	Public receipt #21 SDK module	PASS
D2 Template + quickstart	Copy-paste Core v2 runner using the public SDK: create, sealed commit, Drand reveal, clear, settle/refund, and verified evidence export.	Template and quickstart Evidence pack	PASS
D3 Live demo + 3 receipts	Three real Auction rounds with three bidders each. Every round completed commit -> reveal -> clear -> atomic settle/refund.	#21 #22 #23	PASS

Live Round Evidence

Round	Public receipt / raw JSON	Create transaction	Atomic settlement + refunds
#21 3 bidders 18 XLM winning bid	Receipt page Raw evidence JSON	Open explorer 33c8fdc2...38154aeb	Open explorer 60c0db44...10d9feee
#22 3 bidders 21 XLM winning bid	Receipt page Raw evidence JSON	Open explorer 47c6f5b4...3e8490b4	Open explorer bbc79991...37d972a
#23 3 bidders 22 XLM winning bid	Receipt page Raw evidence JSON	Open explorer 2eb778dc...1eeb5301	Open explorer 161f8c24...72044c32

Atomic settlement note: Core v2 uses one `settle_v2` transaction per round. The same transaction pays the seller, refunds the winner's unused escrow and every losing bidder, and transfers the lot. Therefore the settlement and refund evidence intentionally point to the same transaction hash.

Implementation and Test Evidence

What was delivered

Public receipt surface

The `PublishedAuctionEvidence` SDK model pairs a canonical Core v2 receipt with transaction evidence. Offline verification checks the receipt, Auction/Settled state, bidder minimum, lifecycle hashes, and atomic settlement/refund relationship.

The receipt page reads the current round from Stellar RPC and compares status, mode, bidder count, winner, and winning amount with the published evidence.

Reusable integration template

The template uses native testnet XLM escrow and the public `@sub-rosa/sdk` surface. It creates the round, submits three time-locked bids, fetches the Drand beacon, opens and reveals, clears, settles, and writes verified public JSON.

[Open the copy-paste quickstart](#)

Verification results

Check	Result	Scope
Core v2 contract suite	100 passed	Lifecycle, Drand, policy, settlement, refunds, conservation, error paths
SDK suite	129 passed	Receipt verification, public API, encoding, network and evidence module
Web suite	128 passed	Routes, pilots, configuration, receipt page integration and UI health
Auction template suite	6 passed	Published evidence health plus offline template smoke tests
Production build	Passed	Vite production build and live Vercel deployment
Live ledger reconciliation	3 matched	Rounds #21, #22, and #23 read as Auction / Settled on Stellar testnet

Lifecycle summary

1. Seller escrows the lot and fixes identical 25 XLM bidder escrow.
2. Three allowlisted testnet accounts submit encrypted commitment-bound bids.
3. Amounts remain hidden until Drand round **31393417**.
4. After reveal, the contract records valid bid amounts and clears the highest valid bid.
5. One atomic settlement call pays, refunds all escrow balances, and transfers the lot.
6. The SDK exports a canonical receipt; the public page verifies it offline and reconciles live state.

Known limits and scope boundary

- Testnet only; no real-value or mainnet claim is made.
- The auction lot is a test asset; bidder escrow is native testnet XLM.
- The keeper/runner is demo-grade, not highly available production infrastructure.
- No third-party security audit, production database, multi-tenant dashboard, anti-sybil layer, custom wallet, governance, or fee market is claimed.

SCF separation: Rounds #21-23 and this integration kit were completed as Instawards baseline work before the SCF award. They are not represented as SCF-funded rounds and do not count toward any post-award SCF usage target.

Builder completion declaration

Emin Karagoz

Date: 17 August 2026

Ambassador Chapter Lead review

Signature: _____

Date: _____